CALCULUS II

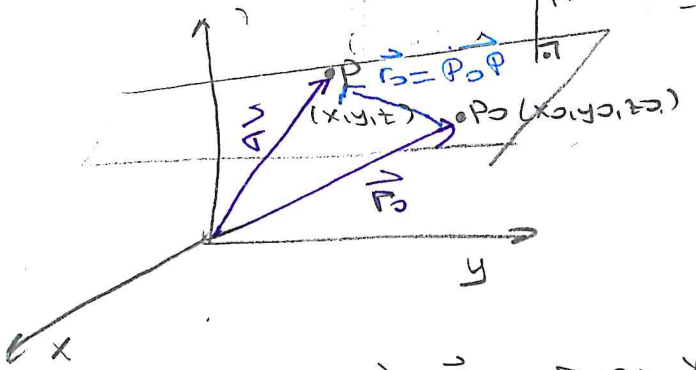
WEEK VI

PLANES IN 3-SPACE

Let $P_0=(x_0,y_0,z_0)$ be a point in R^3 with position vector

$$\vec{r_0}=x_0\vec{i}+y_0\vec{j}+z_0\vec{k}$$

If $\vec{n}=A\vec{i}+B\vec{j}+C\vec{k}$ is any given nonzero vector, then there exists exactly one plane (flat surface) passing through P_0 and perpendicular to $\vec{n}$. We say that $\vec{n}$ is a normal vector to the plane. The plane is the set of all points P for which $\vec{P_0P}$ is perpendicular to $\vec{n}$ (i.e. $\vec{P_0P}\perp\vec{n}$, $\vec{P_0P}\cdot\vec{n}=0$)



The point-normal equation of a plane

$$\vec{n}\cdot(\vec{r}-\vec{r_0})=0$$

$$A(x-x_0)+B(y-y_0)+C(z-z_0)=0$$

$\downarrow$ standard form

$$Ax+By+Cz=D$$

$$D=Ax_0+By_0+Cz_0$$

$P(x,y,z)$ point $\vec{r}$ $\vec{OP}$ position vector

$P_0(x_0,y_0,z_0)$ point $\vec{r_0}$ $\vec{OP_0}$ position vector

Ex (Recognizing and writing the equations of planes)

(a) The equation $2x-3y-4z=0$ represents a plane that passes through the origin and is normal (perpendicular) to the vector $\vec{n}=2\vec{i}-3\vec{j}-4\vec{k}$

(b) Obtain equation of the plane that passes through the point $P_0=(2,0,1)$ and is perpendicular to the straight line passing through the points $(1,1,0)=P_1$ and $(4,-1,-2)=P_2$

$$\vec{n}=\vec{P_2P_1}=(4-1)\vec{i}+(-1-1)\vec{j}+(-2-0)\vec{k}=3\vec{i}-2\vec{j}-2\vec{k}$$

$\vec{n}=(3,-2,-2)$

Normal of the plane is $(3,-2,-2)$ and plane is passing through the point $P_0=(2,0,1)$.

The plane equation is $3(x-2)-2(y-0)-2(z-1)=0$

$$3x-2y-2z=4$$

Show that $2x-y=1$ is parallel to z-axis)

(c) The plane with equations $2x-y=1$ has a normal $2\vec{i}-\vec{j}$ that is perpendicular to z-axis. The normal $(2,-1,0)\perp(0,0,1)$
$[(2,-1,0)\cdot(0,0,1)=0$
of a plane which is parallel to $\vec{k}$ z-axis must be perpendicular to $\vec{k}$. Therefore, $2x-y=1$ is a plane parallel to z-axis.
In the xy-plane, the equation $2x-y=1$ represents a straight line; in 3-space represents a plane containing that line and parallel to z-axis

(d) The equation $2x+y+3z=6$ represents a plane with normal $\vec{n}=2\vec{i}+\vec{j}+3\vec{k}$. Let's find some of its points
For example, $y=z=0$ $x=3$ so $(3,0,0)$ is a point on the plane. We say that the x-intercept of the plane is 3 since $(3,0,0)$ is the point where the plane intersects the x-axis. Similarly, y-intercept is 6,
z-intercept is 2.

(e) In general, if a,b and c are all nonzero, the plane with intercepts a,b and c on the coordinate axes has the equation

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

called the intercept form of the equation of the plane

Ex 7 Find an equation of the plane that passes through the three points $P=(1,1,0)$, $Q=(0,2,1)$ and $R=(3,2,-1)$

We need to find a vector $\vec{n}$ (normal vector of the plane). Since a vector will be perpendicular to the vectors $\vec{PQ}$

$\vec{PQ} = -\vec{i}+\vec{j}+\vec{k}$ $\qquad$ $\vec{PR} = 2\vec{i}+\vec{j}-\vec{k}$.

Therefore, we can use

$$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = -2\vec{i}+\vec{j}-3\vec{k}$$

We can use this normal vector together with the coordinates of any one of the three points to write the equation of the plane. Using point P leads to the equation

$$-2(x-1)+1(y-1)-3(z-0)=0$$

So, $\quad 2x-y+3z=1$

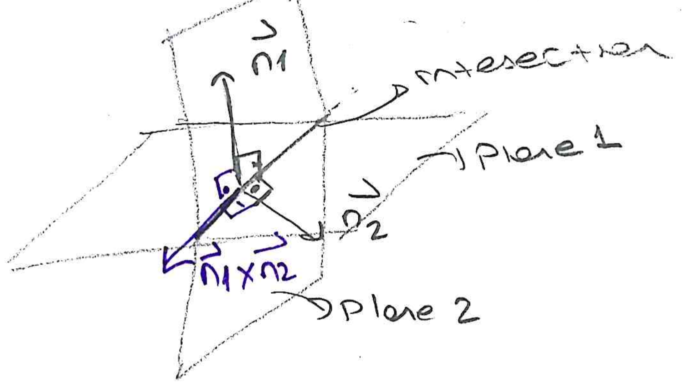
Ex 7 Show that the two planes $x-y=3$ and $x+y+z=0$ intersect, and find a vector, $\vec{v}$, parallel to their line of intersection.

Find normal vectors $\quad x-y=3 \quad \vec{n}_1=(1,-1,0)=\vec{i}-\vec{j}+\vec{k}$

$x+y+z=0 \quad \vec{n}_2=(1,1,1)=\vec{i}+\vec{j}+\vec{k}$

Since these vectors are not parallel the planes are not parallel. ($\vec{n}_1 \neq c\,\vec{n}_2 \quad c\in\mathbb{R}$) and they intersect in a straight line, perpendicular to both $\vec{n}_1$ and $\vec{n}_2$

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = -\vec{i}-\vec{j}+2\vec{k}$$



Please see

$\vec{n}_1 \cdot \vec{n}_2 = 1-1+0=0$

$\vec{n}_1 \perp \vec{n}_2$

Pencil of Planes

A family of planes in a straight line is called a pencil of plane. Such a pencil of planes is determined by any two nonparallel planes in it, since these have a unique line of intersection.

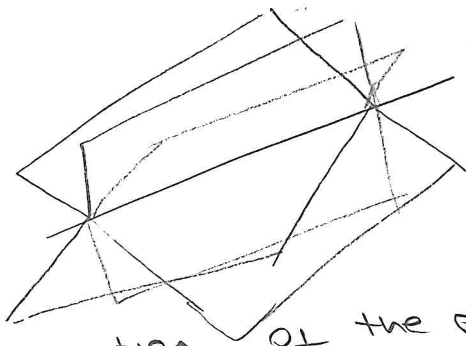
→ If two nonparallel planes have equations

$$A_1x + B_1y + C_1z = D_1 \quad \text{and} \quad A_2x + B_2y + C_2z = D_2$$

then, for any value of the real number λ, the equation

$$A_1x + B_1y + C_1z - D_1 + \lambda(A_2x + B_2y + C_2z - D_2) = 0$$

represent a plane in the pencil



Ex 1 Find an equation of the plane passing through the line of intersection of the two planes

$$x + y - 2z = 6 \quad \text{and} \quad 2x - y + z = 2$$

and also passing through the point $(-2, 0, 1)$

For any constant λ, the equation

$$(x + y - 2z - 6) + \lambda(2x - y + z - 2) = 0$$

represents a plane and is satisfied by the coordinates of all points on the line of intersection of the given planes. This plane will pass through the point $(-2, 0, 1)$, we will find corresponding λ

$$(-2 + 0 - 2 - 6) + \lambda(-4 - 0 + 1 - 2) = 0, \quad \lambda = -2.$$

So, equation of the plane is

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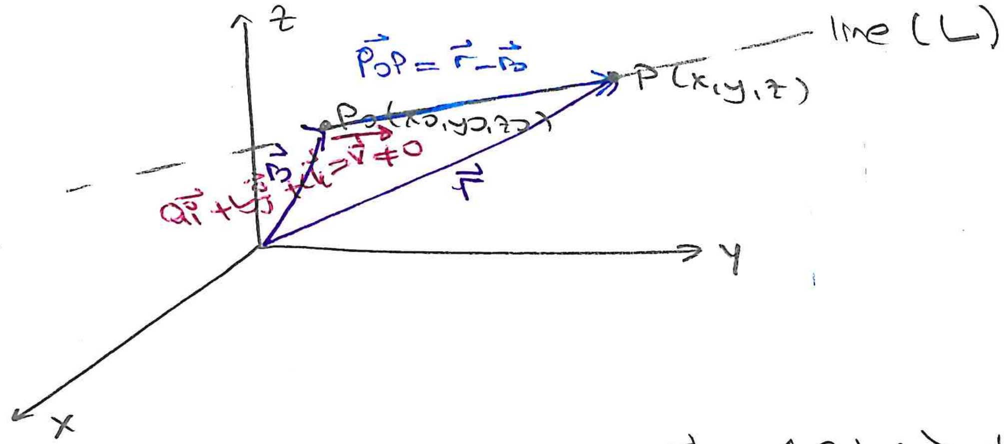
$$\ldots 2) = 3x - 3y + 4z + 2 = \ldots$$

LINES IN 3-SPACE



$\vec{v}$: a vector parallel to L $\quad \vec{v} = \langle a,b,c \rangle \neq \vec{0}$

$P(x,y,z)$: any point on the line

$\vec{r_0} = \overrightarrow{OP_0}$: position vector of point P_0.

$\vec{r} = \overrightarrow{OP}$: position vector of point P

$\overrightarrow{P_0P} = \vec{r} - \vec{r_0}$

Hence, by triangle law $\vec{r} = \vec{r_0} + \overrightarrow{P_0P}$. Note that $\overrightarrow{P_0P}$ and $\vec{v}$ are parallel to each other, therefore, we write $\overrightarrow{P_0P} = t\,\vec{v}$. Then, we get following representation for equations of straight line L

Vector parametric equation

$$\vec{r} = \vec{r_0} + t\vec{v} \qquad -\infty < t < \infty$$

direction vector of the line

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a,b,c \rangle$$

Scalar parametric equation

$$x = x_0 + at \qquad -\infty < t < \infty$$
$$y = y_0 + bt$$
$$z = z_0 + ct$$

Standard form

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

$a,b,c \neq 0$

Remark If for instance $b=0$, we only solve parametric equations for x and z for t. In this case, we have

$$\boxed{\frac{x-x_0}{a} = \frac{z-z_0}{c}, \quad y=y_0}$$

Ex (a) Find equation of straight line passing through the point $(2,3,0)$ and parallel to the vector $\vec{i} - 4\vec{k}$

Standard form $\quad \dfrac{x-2}{1} = \dfrac{z-0}{-4} = t \qquad y=3$

So, parametric form $\quad x = 2+t \qquad -\infty < t < \infty$
$z = -4t$
$y = 3$

(b) Find equation of the straight line passing through $(1,-2,3)$ and perpendicular to the plane $x-2y+4z=5$.

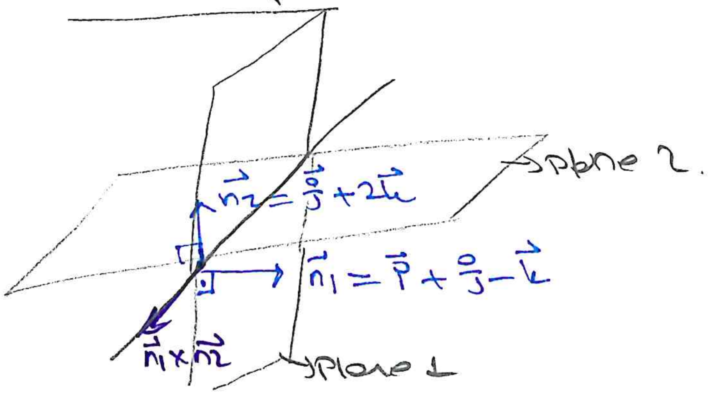
Since line is perpendicular to the plane this means that it is parallel to normal vector of the plane.
$\vec{n} = (1,-2,4)$

Now, we have a line passing through $(1,-2,3)$ and parallel to $\vec{n}$

Standard form $\quad \dfrac{x-1}{1} = \dfrac{y+2}{-2} = \dfrac{z-3}{4}$

Parametric form $\quad x = t+1$
$y = -2t-2$
$z = 3+4t$

(c) Find a direction vector for the line of intersection of the two planes $x+y-z=0$ and $y+2z=6$ and find a set of equations for the line in the standard form



$$\vec{n}_1\times\vec{n}_2=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 0 & 1 & 2\\ 1 & 1 & -1\end{vmatrix}$$

$$=\vec{i}\,(-1-2)-\vec{j}\,(-2+0)+\vec{k}\,(-1+0)$$

$$=-3\vec{i}+2\vec{j}-\vec{k}$$

Now, we know direction of the line $(-3,2,-1)$ we need a point on the line. This point will be intersection point, we will solve $x+y-z=0$ $y+2z=6$ in the mean time take $z=0$ gives $(-6,6,0)$

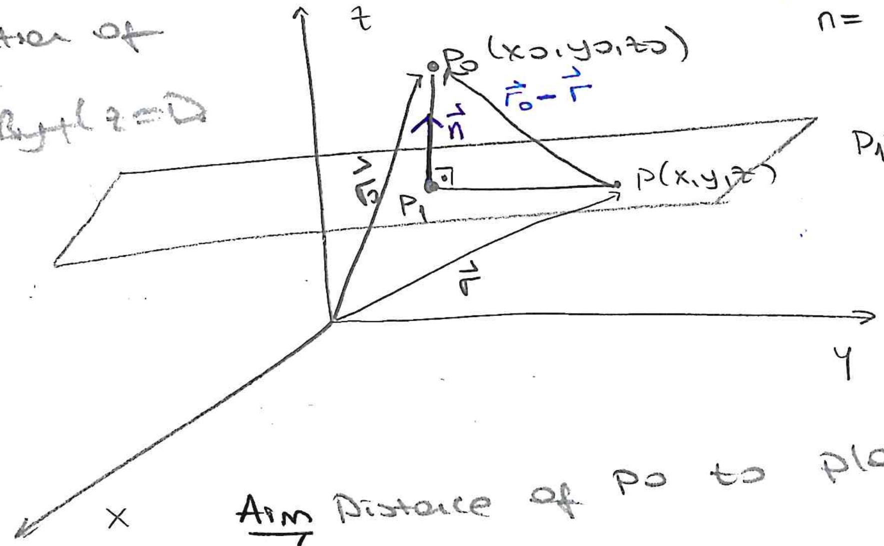
Equation of line passing through $(-6,6,0)$ and parallel to $(-3,2,-1)$ Standard form $\dfrac{x+6}{-3}=\dfrac{y-6}{2}=\dfrac{z-0}{-1}$

Parametric form $x+6=-3t$ $y-6=2t$ $z=-t$

NOTE The vector equation of a line through the vector $\vec{r}_0$ in the direction of $\vec{v}$ is $\vec{r}=\vec{r}_0+t\vec{v}$. If the line passes through r_1 we can take $\vec{v}=r_1-r_0$ and so its vector eqn $\vec{r}(t)=\vec{r}_0+t(\vec{r}_1-\vec{r}_0)=(1-t)\vec{r}_0+t\vec{r}_1$ line segment from r_0 to r_1 $0\le t\le 1$

Distances from a point to a ~~line~~ PLANE

Equation of plane $Ax+By+Cz=D$



$n=A\vec{i}+B\vec{j}+C\vec{k}$: normal to the plane

P_1: the point on the plane closest to P_0.

$\vec{r}_0$: the position vector of P_0.

Aim Distance of P_0 to plane $Ax+By+Cz=D$

→ $\vec{P_1P_0}$ is perpendicular to the plane ⇒ $\overrightarrow{P_1P_0}\parallel\vec{n}$

→ the distance from P_0 to the plane $|\overrightarrow{P_1P_0}|$ = length of the projection of $\overrightarrow{PP_0}=\vec{r}_0-\vec{r}$ in the direction of $\vec{n}$.

$|\overrightarrow{P_1P_0}|$ = length of the projection of $\overrightarrow{PP_0} = \vec{r_0} - \vec{r}$ in the direction of $\vec{n}$ $= \left| \dfrac{\overrightarrow{PP_0} \cdot \vec{n}}{|\vec{n}|} \right| = \dfrac{|(\vec{r_0} - \vec{r}) \cdot \vec{n}|}{|\vec{n}|} = \dfrac{|\vec{r_0}\cdot\vec{n} - \vec{r}\cdot\vec{n}|}{|\vec{n}|}$

Using the fact $\vec{r} \cdot \vec{n} = Ax + By + Cz = D$, we can write

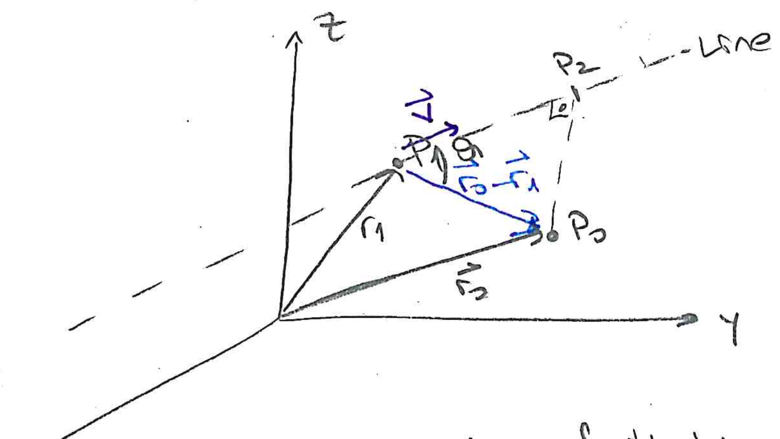
$$\frac{|\vec{r_0}\cdot\vec{n} - \vec{r}\cdot\vec{n}|}{|\vec{n}|} = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

Ex What is the distance from $(2,-1,3)$ to the plane $2x - 2y - z = 9$?

The distance from $(2,-1,3)$ to the plane $2x-2y-z=9$ is

$$s = \frac{|2(2) - 2(-1) - 1(3) - 9|}{\sqrt{(2)^2 + (-1)^2 + (3)^2}} = \frac{|-6|}{3} = 2 \text{ units}$$

Distances from a point to a line



The distance from P_0 to $\mathcal{L}$ $= |\overrightarrow{P_2P_0}|$

$= |\overrightarrow{P_1P_0}| \cdot \sin\theta$

$= |\vec{r_0} - \vec{r_1}| \sin\theta$

$|(\vec{r_0} - \vec{r_1}) \times \vec{v}| = |\vec{r_0} - \vec{r_1}|\,|\vec{v}| \sin\theta$

$$\overset{\uparrow}{=} \frac{|(\vec{r_0} - \vec{r_1}) \times \vec{v}|}{|\vec{v}|}$$

P_2: the point on $\mathcal{L}$ that is closest to P_0.

P_2P_0: perpendicular to $\mathcal{L}$

Ex 7 What is the distance from $(2,0,-3)$ to the line $\vec{r}(t) = \vec{i} + (1+3t)\vec{j} - (3-4t)\vec{k}$

$x=1 \quad \frac{y-1}{3} = \frac{z+3}{4}$

$r(t)$ is parallel to the vector $\vec{v} = 3\vec{j} + 4\vec{k}$

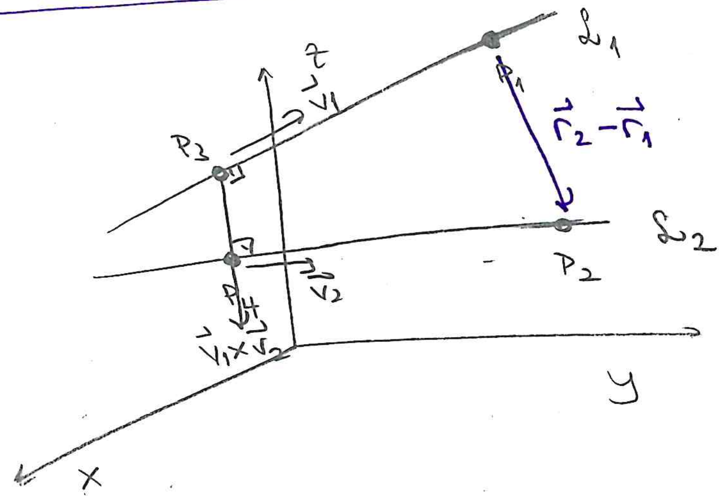
Find a P_1 point on r $\quad P_1 = (1,1,-3) \quad (r(0) = P_1)$,

The distance from $P_0(2,0,-3)$ to this line

$$s = \frac{\left|\left((2-1)\vec{i} + (0-1)\vec{j} + (-3+3)\vec{k}\right) \times (3\vec{j} + 4\vec{k})\right|}{\sqrt{3^2+4^2}}$$

$$= \frac{\left|(\vec{i} - \vec{j}) \times (3\vec{j} + 4\vec{k})\right|}{5} = \frac{\left|-4\vec{i} - 4\vec{j} + 3\vec{k}\right|}{5} = \frac{\sqrt{41}}{5} \text{ unit}$$

The distance between two lines



r_1, r_2: position vectors of P_1, P_2

P_3, P_4: closest points of l_1 and l_2 with position vectors r_3, r_4 respectively

$\overrightarrow{P_3P_4} \parallel \vec{v_1} \times \vec{v_2}$

$\overrightarrow{P_3P_4}$ is the vector projection of $\overrightarrow{P_1P_2}$ along $\vec{v_1} \times \vec{v_2}$

$$|\overrightarrow{P_3P_4}| = \frac{\left|(\vec{r_2} - \vec{r_1}) \cdot (\vec{v_1} \times \vec{v_2})\right|}{|\vec{v_1} \times \vec{v_2}|}$$

EXAMPLES

Ex 1 (a) Find a vector equation and parametric equations for the line that passes through the point $(5,1,3)$ and is parallel to the vector $\vec{i} + 4\vec{j} - 2\vec{k}$

(b) Find two other points on the line

(a) $\vec{r_0} = \langle 5,1,3 \rangle$ position vector of the point $(5,1,3)$

$\vec{v} = \vec{i} + 4\vec{j} - 2\vec{k}$ $\qquad \vec{r} = (5\vec{i} + \vec{j} + 3\vec{k}) + t(\vec{i} + 4\vec{j} - 2\vec{k})$

$$\underbrace{= (5+t)\vec{i} + (1+4t)\vec{j} + (3-2t)\vec{k}}_{\text{vector form}}$$

$$\left.\begin{aligned} x &= 5+t \\ y &= 1+4t \\ z &= 3-2t \end{aligned}\right\} \text{parametric form}$$

(b) By choosing the parameter value t, we can find any points on that line

Let $t=1 \Rightarrow (6,5,1)$ is a point on the line

$t=-1 \Rightarrow (4,-3,5)$ is another point on the line.

Ex 2) Determine if the line that passes through the point $(0,-3,8)$ is parallel to the line given by $x = 10+3t$ $y = 12t$ and $z = -3-t$ passes through the xz plane. If it does give the coordinates of that points.

→ Firstly, write the equation of line. The line is parallel to $x = 10 + \underline{3}t$ $\quad y = \underline{12}t$ $\quad z = -3 - \underline{t}$

that is parallel to $\vec{v} = \langle 3, 12, -1 \rangle$

Hence, the equation of the line is

$$\vec{r} = \langle 0,-3,8 \rangle + t \langle 3,12,-1 \rangle = \langle 3t, -3+12t, 8-t \rangle$$

→ If this line passes through the xz-plane the y coordinate must be zero so $-3+12t=0$ $t=1/4$

→ So, the line passes through the xz-plane at the point

$$\left(3\cdot 1/4,\ -3+12\cdot 1/4,\ 8-1/4\right) = \left(3/4,\ 0,\ 31/4\right)$$

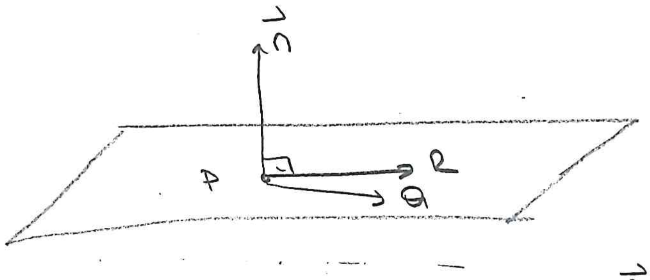
NOTE: The vector equation of a line through the vector $\vec{r}_0$ in the direction of a vector $\vec{v}$ is $\vec{r} = \vec{r}_0 + t\vec{v}$. If the line also passes through r_1 then we can take $\vec{v} = \vec{r}_1 - \vec{r}_0$ so its vector equation is

$$\underbrace{\vec{r} = \vec{r}_0 + t(\vec{r}_1 - \vec{r}_0) = (1-t)\vec{r}_0 + t\vec{r}_1}_{\text{the line segment from } \vec{r}_0 \text{ to } \vec{r}_1}$$

Ex: Determine the equation of the plane that contains the points $P=(1,-2,0)$, $Q=(3,1,4)$, $R=(0,-1,2)$

We first must find the normal vector

$$\vec{PQ} = \langle 2,3,4 \rangle$$
$$\vec{PR} = \langle -1,1,2 \rangle$$



The normal vector can be chosen as $\vec{n} = \vec{PQ} \times \vec{PR}$

$$\vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 4 \\ -1 & 1 & 2 \end{vmatrix} = 2\vec{i} - 8\vec{j} + 5\vec{k} = \langle 2,-8,5 \rangle$$

→ You can use any of the points on the plane to write the equation of the plane. Let us choose the point

$Q=(3,1,4)$

$$2(x-3) - 8(y-1) + 5(z-4) = 0$$
$$2x - 8y + 5z = 18$$

Ex Find the scalar projection and vector projection of

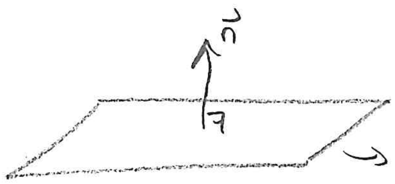
$\vec{b} = \langle 1,1,2 \rangle$ onto $\vec{a} = \langle -2,3,1 \rangle$

$$Proj_{\vec a}(\vec b) = \left(\frac{\vec a \cdot \vec b}{|\vec a|^2} \right) \vec a = \left(scalar_{\vec a} \vec b \right) \cdot \frac{\vec a}{|\vec a|} = \frac{3}{\sqrt{14}} \frac{\langle -2,3,1 \rangle}{\sqrt{14}} = \frac{\langle -6,9,}{\sqrt{14}}$$

$$scalar_{\vec a}(\vec b) = \frac{\vec a \cdot \vec b}{|\vec a|} = \frac{-2 \cdot 1 + 3 \cdot 1 + 1 \cdot 2}{\sqrt{4+9+1}} = \frac{3}{\sqrt{14}}$$

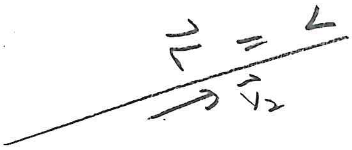
Ex Determine if the plane given by $-x+2z=10$ and the line given by $\vec r = \langle 5, 2-t, 10+4t \rangle$ are orthogonal, parallel or neither

$\vec n = \langle -1, 0, 2 \rangle$ normal vector of the plane



$\to -x+2z=10$

$\vec r = \langle 5, 2-t, 10+4t \rangle = a + t\vec v_2$ $\quad \underbrace{\vec v_2 = \langle 0,-1,4 \rangle}_{\text{is a vector parallel to } \vec r}$



(1) Plane ⊥ Line

Check $\vec n \,//\, \vec v$, this means that plane and line perpendicular to each other. To find it look $\vec n \times \vec v = \vec 0$?

(2) Plane // Line

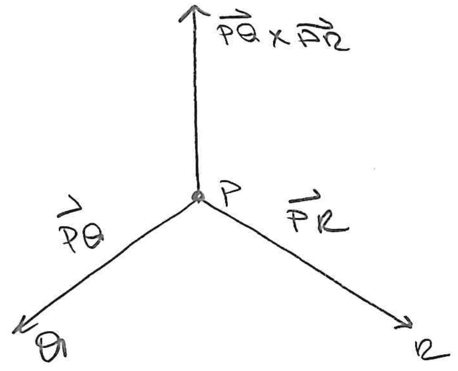
Check $\vec n \perp \vec v$, this means that plane and line will be parallel check $\vec n \cdot \vec v = 0$?

$$\vec n \times \vec v = \begin{vmatrix} \vec i & \vec j & \vec k \\ -1 & 0 & 2 \\ 0 & -1 & 4 \end{vmatrix} = 2\vec i + 4\vec j + \vec k \neq 0$$

Hence, the vector are not parallel and so the plane and line are not ORTHOGONAL

$\vec n \cdot \vec v = 0 + 0 + 8 = 8 \neq 0$ Hence, the vectors are not orthogonal. So, the plane and the line are NOT PARALLEL

Ex 7. Find a vector perpendicular to the plane that passes through $P=(1,4,6)$ $Q=(-2,5,-1)$ $R=(1,-1,1)$



The vector $\vec{PQ}\times\vec{PR}$ is perpendicular to $\vec{PQ}$ and $\vec{PR}$ and is therefore perpendicular to the plane through P, Q and R.

$$\vec{PQ} = \langle -2-1, 5-4, -1-6\rangle = \langle -3,1,-7\rangle$$

$$\vec{PR} = \langle 1-1, -1-4, 1-6\rangle = \langle 0,-5,-5\rangle$$

$$\vec{PQ}\times\vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} = (-5-35)\vec{i} - (15-0)\vec{j} + (15-0)\vec{k} = \langle -40,-15,15\rangle$$

is perpendicular to the given plane. Any nonzero scalar multiple of this vector is also perpendicular to the plane.

Ex 7 Determine if the three vectors $\vec{a}=\langle 1,4,-7\rangle$, $\vec{b}=\langle 2,-1,4\rangle$, $\vec{c}=\langle 0,-9,18\rangle$ lie in the same plane or not (coplanar).

Coplanar: If $|\vec{a}\cdot(\vec{b}\times\vec{c})|=0$, then the vectors must lie in the same plane; that is they are coplanar.

Now, we will check $|\vec{a}\cdot(\vec{b}\times\vec{c})|=0$

$$\vec{b}\times\vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 4 & -7 \\ 2 & -1 & 4 \end{vmatrix} = \vec{i}(-7+16) - \vec{j}(14+4) + \vec{k}(-8-1)$$

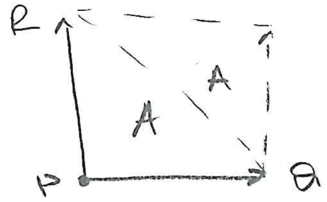
$$= 9\vec{i} - 18\vec{j} - 9\vec{k} = \langle 9,-18,-9\rangle$$

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = \langle 1,4,-7\rangle \cdot \langle 9,-18,-9\rangle$$

$$= 9 - 72 + 63 = 0.$$

$\vec{a},\vec{b},\vec{c}$ lie in the same plane (coplanar)

Ex 7 Find the area of the triangle with vertices $P(1,4,6)$, $Q(-2,5,-1)$ and $R(1,-1,1)$



we know that

$$2A = |\vec{PQ} \times \vec{PR}| = \sqrt{(-40)^2 + (15)^2 + (15)^2}$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} = \vec{i}(-40) - \vec{j}(15) + \vec{k}(15)$$

$\vec{PQ} = (-3, 1, -7)$

$\vec{PR} = (0, -5, -5)$

$$A = \frac{5}{2}\sqrt{82}$$